# Demo

Doing linear regression

# Interpreting Linear Regression

```
Call:
glm(formula = bp_sys ~ bp_dia, data = hrs_df)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 36.42329    2.40097   15.17   <2e-16 ***
bp_dia       1.24770    0.03065   40.71   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’
1

(Dispersion parameter for gaussian family taken to be
229.6316)

    Null deviance: 771374  on 1703  degrees of freedom
Residual deviance: 390833  on 1702  degrees of freedom
AIC: 14103

Number of Fisher Scoring iterations: 2
```

Dependent Variable

Independent Variable

```
Call:
glm(formula = bp_sys ~ bp_dia, data = hrs_df)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 36.42329    2.40097   15.17   <2e-16 ***
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```

Each unit increase in bp_dia results in 1.24770 unit increase in bp_sys.

```
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Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 36.42329    2.40097   15.17   <2e-16 ***
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```

**Null Hypothesis**
Estimate is zero (no relationship).

**Alternative Hypothesis**
Estimate is not zero (there is a relationship.)

# Demo

Multivariable regression

# Knowledge Check

**True or False:** According to the model, being female increases systolic blood pressure.

```
Call:
glm(formula = bp_sys ~ bp_dia + female + age_2014, data =
hrs_df)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -11.53965    5.84565  -1.974   0.0485 *
bp_dia        1.27558    0.02966  43.006  < 2e-16 ***
female       -3.12537    0.75088  -4.162 3.31e-05 ***
age_2014      0.62734    0.06734   9.316  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’
1

(Dispersion parameter for gaussian family taken to be
213.4935)

    Null deviance: 771374  on 1703  degrees of freedom
Residual deviance: 362939  on 1700  degrees of freedom
AIC: 13981

Number of Fisher Scoring iterations: 2
```

**a) True**

**b) False**

# Knowledge Check

**True or False:** According to the model, being female increases systolic blood pressure.

```
Call:
glm(formula = bp_sys ~ bp_dia + female + age_2014, data =
hrs_df)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
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(Dispersion parameter for gaussian family taken to be
213.4935)

    Null deviance: 771374  on 1703  degrees of freedom
Residual deviance: 362939  on 1700  degrees of freedom
AIC: 13981

Number of Fisher Scoring iterations: 2
```

**a) True**

**b) False**

**Is the multivariable model a better fit than the simple model?**

**We can find out with the likelihood ratio test.**

# Likelihood Ratio Test

Compare goodness of fit between two regression models if one is **nested**.

```
Full model:   bp_sys ~ bp_dia + female + age_2014
Nested model: bp_sys ~ bp_dia
```

**H_null:** The full model and the nested model fit equally well.
**H_alternative:** The full model fits better than the nested model.

# Demo

Likelihood Ratio Test

# Knowledge Check

**True or False:** Based on the results of this test, the multivariate model is a better fit.

```
Likelihood ratio test

Model 1: bp_sys ~ bp_dia
Model 2: bp_sys ~ bp_dia + female + age_2014
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   3 -7048.7
2   5 -6985.7  2 126.17  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’
1
```

**a) True**

**b) False**

# Knowledge Check

**True or False:** Based on the results of this test, the multivariate model is a better fit.

```
Likelihood ratio test

Model 1: bp_sys ~ bp_dia
Model 2: bp_sys ~ bp_dia + female + age_2014
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   3 -7048.7
2   5 -6985.7  2 126.17  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’
1
```

**a) True**

**b) False**

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